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(54) Title: **An automated machine-learning system for fetal health prediction using cardiotocography (ctg) data**

(51) International Classification: G16H 50/20, A61B 5/00, A61B 5/024, G06N 20/00 A61B 5/361 (21) Application No: 202641024383 (31) Priority Document No: -- (32) Priority Date: -- (86) International Application No: -- Filing Date: -- (87) International Publication No: -- (61) Patent of Addition to Application No: -- Filing Date: -- (62) Divisional to Application No: -- Filing Date: --	(71) Name of Applicant(s): Vellore Institute Of Technology, (72) Name of Inventor(s): 1. Prabhu J 2. Prakash M 3. Nandhitha H
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(57) Abstract:

The present invention relates to an automated machine-learning system and method for fetal health prediction using cardiotocography (CTG) data are 5 disclosed. the system comprises a data acquisition module configured to receive structured CTG measurements including clinically validated fetal heart rate and uterine contraction features; a data preprocessing module configured to perform exploratory data analysis, normalization, and stratified dataset partitioning while preserving class distribution; and a class-imbalance handling module configured 10 to compute class or sample weights based on inverse-frequency metrics. the system further includes an explainability-driven feature selection module configured to identify clinically significant CTG features, a model learning module configured to train a plurality of machine-learning classifiers using the selected features, and an ensemble stacking module configured to combine 15 outputs of the trained classifiers to generate a final prediction. an explainable artificial intelligence module is provided to generate global and local explanations for prediction decisions, and a prediction and interface module outputs a fetal health classification selected from Normal, Suspect, and Pathological along with associated interpretability information. the disclosed method implements an end-20 to-end automated pipeline that transforms CTG measurements into accurate and interpretable fetal health assessments, thereby supporting clinically informed decision-making. Fig 1 to 3

FORM 2

THE PATENTS ACT, 1970

(39 of 1970)

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THE PATENTS RULES, 2003

COMPLETE SPECIFICATION

(See Section 10; rule 13)

Title of the Invention

**“An automated machine-learning system for fetal health
prediction using cardiotocography (CTG) data”**

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The following specification particularly describes the invention and the manner in which it is performed.

TECHNICAL FIELD

[0001] The present invention relates to the field of medical diagnostics, specifically to automated fetal health monitoring using cardiotocography (CTG) data. it involves machine-learning and explainable AI techniques for predicting
5 and interpreting fetal well-being to support clinical decision-making in prenatal care.

BACKGROUND ART

[0002] Background description includes information that may be useful in understanding the present invention. It is not an admission that any of the
10 information provided herein is prior art or relevant to the presently claimed invention, or that any publication specifically or implicitly referenced is prior art.

[0003] Fetal health monitoring is a critical aspect of prenatal care, aimed at detecting early signs of distress or abnormal development to prevent adverse outcomes. one of the most widely used non-invasive methods for assessing fetal
15 well-being is cardiotocography (CTG), which records fetal heart rate (FHR) and uterine contraction patterns over time. CTG provides valuable clinical indicators such as accelerations, decelerations, short-term variability (STV), long-term variability (LTV), and histogram-derived descriptors that reflect the fetus's physiological state.

20 [0004] Despite the clinical utility of CTG, interpretation of the recordings remains challenging. traditional analysis relies heavily on expert clinicians, leading to variability in diagnosis due to subjective judgment and experience. manual interpretation is also time-consuming and prone to inconsistencies, particularly when large volumes of data must be assessed. furthermore,
25 conventional threshold-based or rule-based CTG assessment methods often fail to capture complex, non-linear relationships between features, limiting predictive accuracy.

[0005] Recent advances in machine learning have shown promise in automating fetal health assessment. machine-learning models can learn complex
30 patterns from structured CTG data and provide more objective predictions. however, existing approaches face several limitations. many models do not

adequately address class imbalance in CTG datasets, leading to reduced accuracy for rare but critical pathological cases. black-box models often lack interpretability, limiting clinical trust and adoption. feature selection is frequently performed without considering clinical relevance, resulting in models that may
5 rely on spurious correlations rather than meaningful physiological indicators.

[0006] EP4174871A1 Techniques are described for performing fetal heart rate (FHR) analytics using machine learning techniques. according to an embodiment, computer-implemented method comprises training a machine learning model using a supervised machine learning process to identify patterns in
10 training cardiotocograph data that correspond to defined physiological events associated with respective fetuses and mothers of the fetuses represented in the training cardiotocograph data. the method further comprises receiving new cardiotocograph data for a fetus and mother in real-time over a period of labor and applying the machine learning model to the new cardiotocograph data as it is
15 received to identify the patterns in the new cardiotocograph data.

[0007] US20240282451A1 One or more systems, devices, computer-implemented methods and/or computer program products of use provided herein relate to artificial intelligence (AI) to generate labor and delivery-based predictions. a system can comprise a processor that can execute computer-executable components stored in memory, wherein the computer-executable
20 components can comprise a first AI model that can generate first data comprising one or more labor and delivery predictions applicable to one or more fetuses and a mother of the one or more fetuses, during labor, by analyzing second data comprising cardiotocography (CTG) analysis data of the one or more
25 fetuses and the mother generated by a second AI model and third data comprising maternal health analysis data of the mother generated by a third AI model, wherein the first AI model can be a multistage AI model comprising respective models directed to predicting respective ones of the one or more labor and delivery predictions.

30 **[0008]** There is therefore a need for a comprehensive system that can automatically process CTG data, handle class imbalance, select clinically relevant

features, and provide accurate and interpretable fetal health predictions. such a system would reduce reliance on manual interpretation, improve diagnostic consistency, and support timely clinical decision-making to enhance fetal outcomes.

- 5 **[0009]** The present invention addresses these challenges by providing an end-to-end automated machine-learning system and method that integrates preprocessing, class-imbalance handling, explainability-driven feature selection, ensemble model learning, and interpretable prediction output, thereby advancing the state of the art in fetal health monitoring.

10 **OBJECTS OF THE INVENTION**

[0010] The principal object of the present invention is to overcome the disadvantages of the prior art.

- [0011]** The primary objective of the present invention is to provide an automated machine-learning system and method for accurate and interpretable fetal health prediction using cardiotocography (CTG) data.

[0012] An objective of the present invention is to develop an end-to-end pipeline that receives structured CTG measurements, preprocesses and normalizes the data, and generates clinically relevant fetal health classifications with minimal manual intervention.

- 20 **[0013]** Another objective of the present invention is to address class imbalance inherent in CTG datasets by computing class or sample weights, thereby improving predictive performance for underrepresented fetal health outcomes.

- [0014]** Another objective of the present invention is to develop an automated machine-learning model that accurately classifies fetal health into Normal, Suspect, and Pathological states using CTG features.

- [0015]** Another objective of the present invention is to employ an explainability-driven feature selection process that identifies clinically significant CTG features, ensuring that model predictions remain interpretable and aligned with clinical knowledge.

[0016] Another objective of the present invention is to integrate BorutaShap-based feature selection for identifying and retaining medically relevant CTG attributes that improve interpretability.

5 [0017] Another objective of the present invention is to utilize ensemble learning and hyperparameter-tuned XGBoost for reliable prediction performance across imbalanced fetal classes.

10 [0018] Another objective of the present invention is to train multiple machine-learning classifiers on the selected features and integrate their outputs using ensemble stacking, enhancing robustness and accuracy of the fetal health predictions.

[0019] Another objective of the present invention is to incorporate SHAP explainability, ensuring transparent and clinically interpretable predictions.

[0020] Another objective of the present invention is to deploy a real-time fetal health prediction interface for clinical and field-level applications.

15 [0021] Another objective of the present invention is to provide global and local explanations of the prediction outcomes using explainable artificial intelligence techniques, allowing clinicians to understand the contribution of each feature and build trust in the system.

[0022] **SUMMARY**

20 [0023] The present invention relates to an automated machine-learning system and method for fetal health prediction using cardiotocography (CTG) data. the invention provides an end-to-end pipeline that transforms structured CTG measurements, including clinically validated fetal heart rate and uterine contraction features, into accurate and interpretable fetal health assessments.

25 [0024] The system comprises a data acquisition module configured to receive CTG data, a preprocessing module for exploratory data analysis, normalization, and stratified partitioning, and a class-imbalance handling module to compute weights that mitigate skewed class distributions. a feature selection module identifies clinically relevant CTG features using explainability-driven importance

comparisons, while a model learning module trains multiple machine-learning classifiers on the selected features. an ensemble stacking module combines the outputs of these classifiers to generate a final fetal health classification.

5 [0025] An explainable artificial intelligence (XAI) module provides global and local explanations of the predictions, enabling clinicians to interpret and trust the system's outputs. the prediction and interface module presents the final fetal health classification normal, suspect, or pathological along with associated interpretability information.

10 [0026] The method of the invention includes receiving CTG data, preprocessing and normalizing it, partitioning it into training and testing sets, addressing class imbalance, selecting significant features, training multiple machine-learning models, generating ensemble predictions, and presenting the outcome with explainability details.

15 [0027] By providing a fully automated, interpretable, and clinically relevant solution for fetal health prediction, the invention enhances early detection of abnormal fetal conditions, supports informed clinical decision-making, and reduces manual workload in prenatal monitoring.

20 [0028] These and other features will become apparent from the following detailed description of illustrative embodiments thereof, which is to be read in connection with the accompanying drawings. While the invention has been described and shown with reference to the preferred embodiment, it will be apparent that variations might be possible that would fall within the scope of the present invention.

BRIEF DESCRIPTION OF DRAWINGS

25 [0029] The accompanying illustrations are incorporated into and form a part of this specification in order to aid in comprehending the current disclosure. The pictures demonstrate exemplary implementations of the current disclosure and, along with the description, assist to clarify its fundamental ideas.

[0030] Fig.1 shows the Proposed framework for Fetal Health Prediction.

[0031] Fig 2 shows the Confusion matrix for fetal health into Normal, Suspect, and Pathological categories.

[0032] Fig 3 shows the Classifying fetal health into Normal, Suspect, and Pathological categories.

5 [0033] It should be noted that the figures are not drawn to scale, and the elements of similar structure and functions are generally represented by like reference numerals for illustrative purposes throughout the figures. It should be noted that the figures do not illustrate every aspect of the described embodiment and do not limit the scope of the present disclosure.

10 [0034] Other objects, advantages, and novel features of the invention will become apparent from the following detailed description of the present embodiment when taken in conjunction with the accompanying drawings.

DETAILED DESCRIPTION OF THE INVENTION

15 [0035] While the present invention is described herein by way of example using embodiments and illustrative drawings, those skilled in the art will recognize that the invention is not limited to the embodiments of drawing or drawings described and are not intended to represent the scale of the various components. Further, some components that may form a part of the invention may not be illustrated in certain figures, for ease of illustration, and such omissions do
20 not limit the embodiments outlined in any way. It should be understood that the drawings and the detailed description thereto are not intended to limit the invention to the particular form disclosed, but on the contrary, the invention is to cover all modifications, equivalents, and alternatives falling within the scope of the present invention as defined by the appended claim.

25 [0036] As used throughout this description, the word "may" is used in a permissive sense (i.e. meaning having the potential to), rather than the mandatory sense, (i.e. meaning must). Further, the words "a" or "an" mean "at least one" and the word "plurality" means "one or more" unless otherwise mentioned. Furthermore, the terminology and phraseology used herein are solely used for
30 descriptive purposes and should not be construed as limiting in scope. Language such as "including," "comprising," "having," "containing," or "involving," and

variations thereof, is intended to be broad and encompass the subject matter listed thereafter, equivalents, and additional subject matter not recited, and is not intended to exclude other additives, components, integers, or steps. Likewise, the term "comprising" is considered synonymous with the terms "including" or "containing" for applicable legal purposes. Any discussion of documents, acts, materials, devices, articles, and the like are included in the specification solely for the purpose of providing a context for the present invention. It is not suggested or represented that any or all these matters form part of the prior art base or were common general knowledge in the field relevant to the present invention.

10 **[0037]** In this disclosure, whenever a composition or an element or a group of elements is preceded with the transitional phrase "comprising", it is understood that we also contemplate the same composition, element, or group of elements with transitional phrases "consisting of", "consisting", "selected from the group of consisting of", "including", or "is" preceding the recitation of the composition, element or group of elements and vice versa.

15 **[0038]** The present invention is described hereinafter by various embodiments with reference to the accompanying drawing, wherein reference numerals used in the accompanying drawing correspond to the like elements throughout the description. This invention may, however, be embodied in many different forms and should not be construed as limited to the embodiment set forth herein. Rather, the embodiment is provided so that this disclosure will be thorough and complete and will fully convey the scope of the invention to those skilled in the art. In the following detailed description, numeric values and ranges are provided for various aspects of the implementations described. These values and ranges are to be
20 treated as examples only and are not intended to limit the scope of the claims. In addition, several materials are identified as suitable for various facets of the implementations.

25 **[0039]** The present invention relates to an automated machine-learning system and method for fetal health prediction using cardiotocography (CTG) data. The system is designed to provide accurate, explainable, and clinically relevant assessments of fetal health by processing CTG measurements in an end-to-end
30

automated pipeline.

[0040] In an embodiment of the present invention, an automated machine-learning system for fetal health prediction using cardiotocography (CTG) data, comprises: i) a data acquisition module configured to receive structured CTG data comprising a plurality of clinically validated fetal heart rate and uterine contraction features; ii) a data preprocessing module configured to perform exploratory data analysis, normalization, and stratified dataset partitioning while preserving class distribution; iii) a class-imbalance handling module configured to compute class or sample weights based on inverse-frequency metrics; iv) a feature selection module configured to identify clinically relevant CTG features using explainability-driven importance comparison; v) a model learning module configured to train multiple machine-learning classifiers using the selected features; vi) an ensemble stacking module configured to combine outputs of a plurality of trained classifiers to generate a final prediction; vii) an explainable artificial intelligence module configured to generate global and local explanations for prediction decisions; and viii) a prediction and interface module configured to output a fetal health classification selected from Normal, Suspect, and Pathological, wherein the system operates as an end-to-end automated pipeline transforming CTG measurements into interpretable fetal health assessments.

[0041] In another embodiment of the present invention, the method for automated fetal health prediction using CTG data, comprises: i) receiving CTG data comprising a predefined set of fetal heart rate and uterine contraction features; ii) performing exploratory data analysis and normalization of the CTG data; iii) partitioning the CTG data into training and testing subsets using stratified sampling; iv) computing class weights to compensate for class imbalance; v) selecting a subset of clinically significant CTG features using an explainability-based feature selection process; vi) training a plurality of machine-learning models using the selected features; vii) generating a final fetal health classification by stacking ensemble predictions; and viii) presenting the classification outcome with associated explainability information.

[0042] In an aspect of the present invention, the invention functions as an automated machine-learning-driven fetal health prediction system designed to classify fetal conditions into Normal, Suspect, and Pathological categories using cardiotocography (CTG) data. it operates through a structured pipeline beginning
5 with the acquisition of 21 clinically validated CTG features, including accelerations, decelerations, short-term variability, long-term variability, and histogram-based measures that collectively represent fetal physiological behavior. since CTG signals naturally exhibit variability and occasional noise, the system first performs extensive exploratory data analysis, generating distribution plots,
10 correlation matrices, outlier maps, and automated profiling reports to understand feature behavior and detect abnormalities.

[0043] In another aspect of the present invention, all features are normalized using MinMax scaling to maintain uniformity across the dataset, and a stratified train-test split ensures that the inherent class imbalance dominated by Normal
15 cases is preserved during model development. to address this imbalance, the system computes class weights and sample weights based on inverse-frequency metrics, ensuring that minority classes such as suspect and pathological receive proportional learning emphasis.

[0044] In another aspect of the present invention, a key component of the
20 invention is its integration of BorutaShap feature selection, which uses SHAP values and iterative shadow-feature comparisons to identify the most clinically relevant predictors. this process reduces the original 21 features to approximately 11 core features, each verified to have consistent medical significance, thereby improving model interpretability and reducing overfitting risks.

[0045] In another aspect of the present invention, using these selected
25 features, the system develops multiple baseline machine-learning models, including logistic regression, gaussian naïve bayes, support vector classifier, KNN, decision tree, random forest, gradient boosting, and XGBoost. these models are evaluated using stratified cross-validation, with F1-macro score as the primary
30 metric due to the imbalanced nature of the data. XGBoost emerges as the optimal

model owing to its ability to capture nonlinear interactions, handle imbalanced data via built-in weighting, and maintain performance stability across multiple training cycles.

5 **[0046]** In another aspect of the present invention, to further enhance predictive performance, an ensemble stacking mechanism is applied, combining random forest, gradient boosting, and XGBoost as base models, with a random forest meta-learner aggregating predictions. this stacked architecture minimizes bias and variance errors, significantly improving classification consistency for borderline suspect and pathological cases.

10 **[0047]** In another aspect of the present invention, the system then undergoes hyperparameter tuning using GridSearchCV, optimizing parameters such as maximum depth, number of estimators, learning rate, and regularization constraints. these improvements result in enhanced predictive metrics, including accuracy values approaching 95% and weighted F1-scores nearing 97%.

15 **[0048]** In another aspect of the present invention, a distinctive feature of the invention is its built-in explainable AI module using SHAP visualizations. global SHAP summaries reveal the most influential CTG features contributing to predictions, while local SHAP force plots show how specific values push individual cases toward different risk categories, ensuring transparency and
20 clinical interpretability.

[0049] In another aspect of the present invention, the complete invention is deployed through an interactive Streamlit-based web interface that enables both real-time single-record predictions and batch analysis through CSV uploads. users enter CTG parameters or upload multiple samples, after which the system
25 classifies each case and, when required, displays supporting visualizations such as confusion matrices and SHAP impact diagrams. the interface provides colorcoded risk indicators, making it suitable for rapid screening in clinical and remote healthcare environments.

[0050] In another aspect of the present invention, internally, the architecture

integrates modules for preprocessing, feature selection, model training, stacking ensemble computation, explainability analysis, and final prediction output, all arranged in a modular schema that supports scalability and future integration with electronic health record systems. overall, the invention presents a unified solution that transforms raw CTG measurements into reliable, interpretable, and clinically actionable fetal health assessments through an intelligent combination of machine learning, advanced feature selection, ensemble modeling, and explainable artificial intelligence.

[0051] In another aspect of the present invention, the table 1 shows traditional statistical models (logistic regression, GaussianNB) performed reasonably but lacked the ability to model complex nonlinear interactions present in CTG data. SVM and decision tree models showed moderate performance but were more sensitive to noise and imbalance. the XGBoost and stacking ensemble significantly outperformed baseline models in terms of F1-macro and recall, especially for minority classes. compared to simple models, XGBoost's gradient boosting mechanism provided better handling of medical data variability and improved prediction stability.

[0052] Table 1. Baseline models

Model Name	Train Accuracy Mean	Test Accuracy Mean	Fit Time Mean
Random Forest Classifier	0.999159	0.896338	0.298136
Logistic Regression	0.746803	0.734529	0.066032
SVC	0.674599	0.661816	0.263828
Logistic Regression	0.787592	0.767417	0.630440
Gradient Boosting Classifier	0.993484	0.913415	1.317132
Gaussian NB	0.704489	0.700913	0.007825
XGB Classifier	0.999158	0.913334	1.366641

[0053] In another aspect of the present invention, the optimized XGBoost

model demonstrated high predictive accuracy in classifying fetal health into Normal, Suspect, and Pathological categories using CTG features as shown in figure 2 and figure 3. by combining feature selection (BorutaShap) with ensemble learning, the system successfully captured complex physiological patterns and variability in fetal heart rate signals. the model performed consistently across crossvalidation folds, showing strong generalization ability despite class imbalance. SHAP interpretability confirmed that the model focused on clinically valid indicators such as short-term variability, accelerations, and deceleration patterns.

10 **[0054]** Further, the operations need not be performed in the disclosed order, although in some examples, an order may be preferred. Also, not all functions need to be performed to achieve the desired advantages of the disclosed system and method, and therefore not all functions are required.

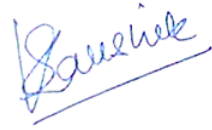
15 **[0055]** Various modifications to these embodiments are apparent to those skilled in the art from the description and the accompanying drawings. The principles associated with the various embodiments described herein may be applied to other embodiments. Therefore, the description is not intended to be limited to the embodiments shown along with the accompanying drawings but is to be providing the broadest scope consistent with the principles and the novel and
20 inventive features disclosed or suggested herein. Accordingly, the invention is anticipated to hold on to all other such alternatives, modifications, and variations that fall within the scope of the present invention and appended claims.

CLAIMS

I/We Claim:

1. An automated machine-learning system for fetal health prediction using cardiotocography (CTG) data, comprising:
 - 5 i) a data acquisition module configured to receive structured CTG data comprising a plurality of clinically validated fetal heart rate and uterine contraction features;
 - ii) a data preprocessing module configured to perform exploratory data analysis, normalization, and stratified dataset partitioning while preserving
10 class distribution;
 - iii) a class-imbalance handling module configured to compute class or sample weights based on inverse-frequency metrics;
 - iv) a feature selection module configured to identify clinically relevant CTG features using explainability-driven importance comparison;
 - 15 v) a model learning module configured to train multiple machine-learning classifiers using the selected features;
 - vi) an ensemble stacking module configured to combine outputs of a plurality of trained classifiers to generate a final prediction;
 - vii) an explainable artificial intelligence module configured to generate
20 global and local explanations for prediction decisions; and
 - viii) a prediction and interface module configured to output a fetal health classification selected from Normal, Suspect, and Pathological, wherein the system operates as an end-to-end automated pipeline transforming CTG measurements into interpretable fetal health assessments.
- 25 2. The system as claimed in claim 1, wherein the CTG data comprises clinically validated features, including accelerations, decelerations, short-term variability, long-term variability, and histogram-derived descriptors.

3. The system as claimed in claim 1, wherein the feature selection module employs BorutaShap feature selection using SHAP-based importance comparison with shadow features.
4. The system as claimed in claim 1, wherein the model learning module trains at
5 least one gradient-boosting-based classifier.
5. The system as claimed in claim 1, wherein the explainable artificial intelligence module generates global feature-importance summaries and local instance-level explanations using SHAP-based attribution.
6. The method for automated fetal health prediction using CTG data, as claimed in
10 claim 1, comprising:
- i) receiving CTG data comprising a predefined set of fetal heart rate and uterine contraction features;
 - ii) performing exploratory data analysis and normalization of the CTG data;
 - 15 iii) partitioning the CTG data into training and testing subsets using stratified sampling;
 - iv) computing class weights to compensate for class imbalance;
 - v) selecting a subset of clinically significant CTG features using an explainability-based feature selection process;
 - 20 vi) training a plurality of machine-learning models using the selected features;
 - vii) generating a final fetal health classification by stacking ensemble predictions; and
 - viii) presenting the classification outcome with associated explainability
25 information.
7. The method as claimed in claim 6, wherein normalization is performed using MinMax scaling to map numerical features to a uniform range.



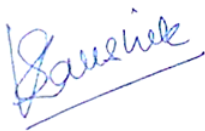
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ABSTRACT

“An automated machine-learning system for fetal health prediction using cardiotocography (CTG) data”

The present invention relates to an automated machine-learning system and method for fetal health prediction using cardiotocography (CTG) data are disclosed. the system comprises a data acquisition module configured to receive structured CTG measurements including clinically validated fetal heart rate and uterine contraction features; a data preprocessing module configured to perform exploratory data analysis, normalization, and stratified dataset partitioning while preserving class distribution; and a class-imbalance handling module configured to compute class or sample weights based on inverse-frequency metrics. the system further includes an explainability-driven feature selection module configured to identify clinically significant CTG features, a model learning module configured to train a plurality of machine-learning classifiers using the selected features, and an ensemble stacking module configured to combine outputs of the trained classifiers to generate a final prediction. an explainable artificial intelligence module is provided to generate global and local explanations for prediction decisions, and a prediction and interface module outputs a fetal health classification selected from Normal, Suspect, and Pathological along with associated interpretability information. the disclosed method implements an end-to-end automated pipeline that transforms CTG measurements into accurate and interpretable fetal health assessments, thereby supporting clinically informed decision-making.

Fig 1 to 3.



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